Sahaj Bhadja

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Summary

Detail-oriented quality assurance engineer experienced in **Python automation, command-line scripts, test execution, configuration files, defect investigation, ML data validation, and reporting**, improving software quality across collaborative teams.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Tested Python services processing 5,000+ monthly jobs using input validation, reconciliation checks, error handling, and state tracking, increasing concurrent throughput 3x while reliably sustaining 99%+ production reliability.
- Investigated performance and data-consistency defects for 500+ users, clearly documenting findings and improving dashboard latency from 1.2 seconds to 350 milliseconds through indexing, caching, and query optimization.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Executed automated tests and validated 1,000+ PDF, DOCX, ZIP, text, and image inputs per batch, reliably identifying ingestion, OCR, model, timeout, and API failures under concurrent workloads.
- Implemented Python diagnostics, fallback behavior, model-usage logs, and structured test data collection across 10,000+ daily requests, reliably reducing feedback turnaround time 60% while maintaining stable peak-load performance.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible Python and PyTorch data pipelines processing 100,000+ trajectory records, validating model behavior through simulation, edge-case analysis, debugging, and structured experiments while improving planning accuracy 18%.
- Monitored closed-loop interactions across ego, road, and nearby-agent signals, clearly documenting test findings and implementation decisions while improving system stability 15% through methodical analysis for senior researchers.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Reliably maintained automated test scripts covering executable builds, installations, and 20+ unit and integration cases across Windows, macOS, and Linux using Jenkins, PyTest, Mockito, Shell, and Bash.
- Investigated defects through failure logs, CSV reports, and code-coverage summaries, improving API response times 30%, increasing nightly pipeline efficiency 68%, and reducing technical debt 20% across releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Created and executed unit, integration, and end-to-end tests for eight services and two million synchronized records, achieving 95%+ successful build validation across production release cycles through Jenkins-based quality checks.
- Monitored errors, performance, logs, and usage through Sentry, Azure Monitor, Kibana, and Google Analytics, translating identified issues into improvements that reduced resource wastage 42% across production workflows.

Skills

QA: Test cases, test scripts, test execution, test data, defect investigation, failure reporting, quality assurance
Automation: Python, Shell, Bash, command line, configuration files, PyTest, Mockito, Jenkins, CI/CD
Testing: Unit testing, integration testing, end-to-end testing, reproducible testing, validation, edge-case testing
Tools: Git, Linux, Windows, macOS, Docker, Sentry, Azure Monitor, Kibana, logging, monitoring

Projects

CPU-scheduling Simulator Live | *Automated Functional Testing*
- Implemented and tested six C++ scheduling algorithms with edge-case handling, automated metrics, and Gantt visualization, executing 1,000+ process combinations while reducing average turnaround time 25% and debugging time 20%.

CaptionCraft - Image Caption Generator Live | *ML Data Validation*
- Collected and evaluated results across 10,000 images and 100,000 captions, benchmarking four model architectures through repeatable experiments and improving measured caption accuracy from 81% to 96% at scale.

Comprehensive Analysis of Blockchain for User Authentication | *IoT Security Testing*
- Evaluated IoT authentication behavior across three user-access scenarios and four permission tiers, identifying security edge cases and designing tamper-resistant workflows that reduced repeated key-verification steps by 50%.